A272 – Carrier and Information

Johann Pascher

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Abstract. The familiar claim that "erasing a bit produces at least $kT \ln 2$ of heat" goes beyond what Landauer's 1961 paper proves. Landauer shows something precise: if a physical two-state system in thermal equilibrium is forced into a definite state irrespective of its initial state, at least $kT \ln 2$ must be dissipated. That is a statement about *carrier operations*. The step from there to a universal claim about *information* as such is not taken in Landauer's paper; it is encouraged by a conceptual ambiguity in the word "bit", but it is not a proved consequence of his arguments. This paper separates the two, documents Landauer's own restriction, traces the generalisation through the reception literature, and names the linguistic mechanism — the hypostatisation of an abstraction — that carried it. *Relation to A271:* A272 adds nothing to the physics. A271 does the accounting; A272 does the textual criticism. The dictionary mapping the two vocabularies is given in Section .1.

Layer markings. **[SETZUNG]** declared starting point (German corpus token: [SETZUNG]) — **[B]** derived algebraically/mathematically from declared foundations — **[Q]** source — external primary source or measured value — **[S]** plausibility sketch — **[H]** open research question.

.1 The question at issue

Since Landauer's 1961 paper [B1] it has often been claimed in the literature that every erased bit must produce at least $kT \ln 2$ of heat. The claim has become a fixture of computer science, physics and the philosophy of science.

The present paper holds that this generalisation goes beyond what Landauer's proof shows. Landauer makes a precise statement about physical carrier operations. The jump from there to a universal statement about information as an abstract quantity is not his step — it was taken in the reception, encouraged by a conceptual imprecision that is built into the discipline.

Two delimitations first, so that the reach of this paper is not overestimated.

[B] First: this is not a refutation of Landauer's principle. What is contested here is the *popular short form*, not Landauer's calculation. For the physical operation Landauer considers, his result is correct. What is contested is the jurisdiction of that result over a subject matter — abstract information — about which it makes no statement.

[B] Second: the physical substance is in A271, not here. That the thermodynamic lower bound attaches to the physical region and not to its description is derived there from Liouville. A272 adds nothing to that. What A272 does is different and stands on its own: the textual analysis of Landauer 1961, the documentation of Landauer's own restriction, the reception history of the generalisation, and the critique of the language of digital technology.

Dictionary: A271 and A272

[SETZUNG] The two documents draw the same distinction in different words. So that the corpus does not end up carrying two parallel terminologies for one matter, the mapping is fixed here and is binding:

A271 (accounting)	A272 (textual criticism)	Level
region in phase space	carrier	physical
description	information	abstract
region reduction $\{A, B\} \to Z$	carrier operation RESTORE TO ONE	physical
non-bijectivity of the region map	shrinking of the carrier state space	physical
descriptive decision	interpretive erasure	abstract

[B] The mapping is a translation, not an extension. Whoever reads "carrier" may think "region", and conversely. A statement that holds in one column holds in the other — otherwise there is an error in one of the two documents, not a new matter of fact.

.2 The core distinction

Definitions

Definition .2.1 (Information)**.** Information is an abstract quantity. A bit is a unit of information representing one of two states (0 or 1). Information has no mass, no energy, no entropy and no physical location.

Definition .2.2 (Carrier)**.** The carrier of the information is a physical system that represents the information: an electrical circuit, a magnetic spin, a photon or any other physical two-state system. The carrier has energy, entropy and further physical properties.

[SETZUNG] Both definitions are stipulations. They are not proved; they fix what is being talked about in what follows. The entire yield of the paper depends on keeping the separation intact.

Energy attaches to the carrier

Theorem .2.3. *No energy can be assigned to a bit of information, only to its carrier.*

[B] Proof. Suppose a bit of information had an intrinsic energy. Then that energy would have to be independent of the physical realisation. But a bit with the value 1 can be represented by different physical systems with different energies: by 5 V in a CMOS circuit, by 1 V in another technology, by a magnetic spin with a given exchange energy, by a photon of a given frequency. Since the information content is identical in all cases while the energy varies, the energy cannot be a property of the information. It is a property of the carrier. □

The argument is a multiple-realisability argument and carries exactly as far as multiple realisability reaches — that is, all the way: there is no class of realisations in which the energy would co-vary with the information content.

The bus example

[B] An instructive example is the bit width of a data bus:

- The **bus width** (say 64 bits) is a fixed physical property of the hardware. It determines how many carriers are switched simultaneously.
- The **information** is what is encoded by the patterns. The same bit pattern can mean different things in different contexts.

A 64-bit bus always switches 64 carriers, regardless of whether the information is a number, a letter or a picture element. The lines have a certain capacitance and require energy for recharging. The information does not have that energy.

.3 Landauer's model — what it shows

The particle in the double well

[Q] Landauer's central model (Fig. 1 in [B1]) is a classical particle in a bistable potential well:

- left minimum: state 0;
- right minimum: state 1;
- the operation RESTORE TO ONE forces the particle into the right well irrespective of its initial state.

This is a classical macroscopic model. It does not capture the full breadth of modern quantum-information concepts, but it remains valid as an analogy for bistable systems — spin-½ systems are formally equivalent. What matters is what the model does *not* do: it shows under which physical conditions dissipation is unavoidable. It does *not* show that those conditions are met whenever a bit is erased in the information-theoretic sense.

What Landauer's model does	What Landauer's model does not do
proof for physical carrier operations	proof for information-theoretic erasure
lower bound for RESTORE TO ONE	statement about abstract bits
validity for a thermalised ensemble	validity for arbitrary computational processes
physical entropy $S = k \ln W$	bridge to Shannon entropy

The entropy calculation

[Q] Landauer computes the entropy change on resetting as

$$\Delta S \;=\; k \ln 2 - k \ln 1 \;=\; k \ln 2. \tag{1}$$

This calculation presupposes:

1. **before:** the carrier can take 2 physical states ($W = 2$);
2. **after:** the carrier can take only 1 physical state ($W = 1$).

Both presuppositions are statements about the carrier. Neither is a statement about the information content.

.4 The objection and its reach

[B] The entropy change in equation (1) is a property of the **physical carrier**, not of the information. If a bit is erased *interpretively* — by redefining its meaning, declaring the value invalid, striking the address from a table — then the number of physical states of the carrier does **not** change. It still holds that $W = 2$, hence

$$\Delta S = k \ln 2 - k \ln 2 = 0. \tag{2}$$

Theorem .4.1. *Landauer's principle bites only if the operation physically shrinks the state space of the carrier. A purely interpretive erasure does not shrink it.*

[B] Proof. Let W be the number of physical microstates of the carrier. An interpretive erasure declares one state no longer valid; it is an operation at the level of description and leaves the physical state space unchanged. The entropy of the carrier remains $S = k \ln W$. Only when the carrier itself is forced into a single state ($W' = 1$) does the entropy change to $S' = 0$. That is a physical operation on the carrier, not on the information content. □

What this objection is not directed against

[B] Precision is needed here, otherwise the objection is aimed at an opponent who does not exist. Equation (2) does *not* refute Landauer. Landauer nowhere claims that a conceptual redesignation costs heat; nobody in the technical literature claims that. What equation (2) shows is solely that the two processes — interpretive erasure and carrier reset — come apart, and that only the second falls under Landauer's bound.

That is not a small thing, but it is also not what it looks like at first glance. The yield is a *finding about jurisdiction*, not a refutation:

> **[B]** Landauer's bound has jurisdiction over carrier operations that shrink the state space. For everything else that ordinary language calls "erasing", it has no jurisdiction — neither confirming nor refuting.

[B] The real question thereby shifts, and it is harder than the original one: *does information-theoretic erasure enforce a state-space-shrinking carrier operation?* Bennett [B2][B8] showed that in general it does not — any logically irreversible function can be executed reversibly if the intermediate states are carried along. Only the final relinquishing of carrier states costs. So the answer for the general case is negative, and it is an answer Landauer could not yet give in 1961.

.5 Landauer's own restriction

[Q] Landauer formulated a restriction that the later reception mostly passes over:

> *"Our reset operation, in the preceding discussion, was applied to a thermal equilibrium ensemble. In actuality we would like to know what happens in a particular computing circuit which will work on information which has not yet been thermalized ..."* [B1]

[Q] Landauer resolves the question himself immediately afterwards: for a random sequence of ones and zeros he equates the statistical ensemble with the time average and thereby arrives at the same dissipation per reset as for the thermalised ensemble. For non-random data — if, say, all inputs are already ONE — he expressly concedes that no entropy change occurs.

Three things follow:

1. **[Q]** Landauer is aware of the limits of his model and works them out himself. The charge of over-generalisation hits the reception, not him.
2. **[B]** His result $kT\ln 2$ holds strictly for the equilibrium ensemble. For correlated, non-random data, $kT\ln 2$ per carrier is an *upper* bound; the applicable bound is kTH, with H the entropy of the source in nats. This is not a weakening of the principle but its correct statement: the bound follows the actual state count, not the bit width.
3. **[B]** Above all: this extension too remains a statement about *carrier states*. The transition to abstract information is not made here either.

.6 What Landauer proves — and what he does not

The popular short form reads:

> *"Every erased bit produces $kT\ln 2$ of heat."*

[B] This claim goes beyond Landauer's proof. What is proved is:

> *If a physical two-state system in thermal equilibrium is forced into a definite state irrespective of its initial state, at least $kT\ln 2$ of heat must be released. This dissipation is a property of the physical carrier operation.*

[Q] The step Landauer's proof does not contain is the bridge from the carrier operation to abstract information. Landauer uses "bit" throughout in a double sense — sometimes for the physical object, sometimes for the unit of information. This ambiguity is present in the original paper and encouraged the later reception, in which a statement about carriers became a statement about information.

[B] That every information-theoretic erasure requires a state-space-shrinking carrier operation is presupposed, not derived — and by Bennett it is in general false.

.7 The modern thermodynamics of information

[Q] The thermodynamics of information with feedback [B5][B6] gives, for the erasure process, the generalised bound

$$Q_{\text{min}} = kT(\ln 2 - I), \qquad I \in [0,\ \ln 2], \tag{3}$$

where I is the mutual information between measuring device and system state, measured in **nats** and therefore dimensionless. Equivalently in the form of Sagawa and Ueda: $W_{\text{ext}} \leq -\Delta F + kTI$.

- **[B]** If the experimenter knows whether the carrier is at 0 or 1 ($I = \ln 2$), the carrier can be reset reversibly — $Q_{\text{min}} = 0$.
- **[B]** If not ($I = 0$), the reset must be irreversible — $Q_{\text{min}} = kT \ln 2$.

[B] The heat therefore arises not *because* information is lost, but because of the lack of knowledge about the carrier state. The information is the label stuck onto the carrier; the price attaches to how far the label actually pins the carrier down.

[B] And the symmetry must be seen at the same time: the knowledge I is itself stored on a carrier. Whoever uses it to reset reversibly has not avoided the price but displaced it — to the later moment at which the carrier of the knowledge is itself reset. That is Bennett's resolution of Maxwell's demon [B8], and it prevents equation (3) from being read as a circumvention of the bound.

.8 The perpetuation in the reception

[Q] The conceptual imprecision would have remained inconsequential had the subsequent literature corrected it. The opposite happened. The majority of publications after 1961 adopt the principle in generalised form without going back to Landauer's own restrictions.

- **[Q] Lairez** [B9] notes that criticism of the generalisation has been available for decades but has had no broad effect.
- **[Q] Norton** [B4] shows that the standard argument rests on an untenable identification of dynamic and static probabilities; **Earman and Norton** [B3] had already worked through the line from Szilárd to Landauer in 1999.
- **[Q] Hemmo and Shenker** [B10] show that in the light of the multiple-computations theorem the principle cannot be universally valid.

[S] The reason for the persistence lies not in a refutation of this criticism but in structural features of scientific communication: the formula $kT \ln 2$ is catchy and easy to cite. The 1961 original is rarely read; what gets cited are secondary sources citing secondary sources. Trust in the authority of a distinguished industrial researcher does the rest. The consequence is that a step never taken in the proof is traded as a "fundamental law of nature" — a description that does not appear in the original paper.

[B] This diagnosis is a plausibility sketch and is marked as such. It names mechanisms that are individually observable but whose combined effect is not quantified. A defensible version would be a citation analysis: the share of papers citing [B1] that also state the ensemble restriction. That analysis is not provided here.

.9 The imprecision in the language of digital technology

[B] The problem is not confined to Landauer. The whole of digital technology uses a language that systematically states carrier properties as properties of information. The practice is so deeply entrenched that practitioners rarely perceive it as problematic.

Customary expression	Physical reality (carrier)	Problematic implication
"set a bit"	apply a voltage to a capacitor	information has a state
"erase a bit"	set the carrier to a reference voltage	information can be annihilated
"transmit a bit"	reconstruct a signal on a new line	information is transportable

Customary expression	Physical reality (carrier)	Problematic implication
"copy a bit"	read a voltage, generate it elsewhere	information multiplies itself
"the RAM holds 8 GB"	8 billion physical memory cells	information is a bulk quantity
"bit rate"	voltage changes per second	amount of information per unit time

In each case a property of the carrier is phrased as if it were a property of the information. In practice engineers know that they are handling voltages, currents and charges. In the descriptive language, however, the bit becomes the acting subject: it "flows through the bus", "is stored", "is erased". This language is convenient for many purposes — but it tempts one to apply physical laws directly to the abstraction, as the popular Landauer reception did.

[B] A more precise usage would distinguish:

- **Not:** "the bit has energy." **But:** "the carrier of this bit has energy."
- **Not:** "the bit is erased." **But:** "the carrier is set to a definite state."
- **Not:** "the bit costs $kT\ln 2$." **But:** "this carrier operation costs at least $kT\ln 2$ under these conditions."

The connection to Landauer is direct: he uses "bit" in exactly this double sense. The ambiguity is not his invention; it is built into the discipline and it carried the generalisation.

.10 Epistemological background: hypostatisation

[S] The mechanism described — an abstraction is treated as a physical quantity and equipped with physical properties — has a name: *hypostatisation* (or reification), the transformation of an abstract concept into a supposedly real, substantial object.

[Q] The bit is an abstraction. Shannon [B11] defined it as a statistical property of signals, not as a physical object. Wiener stated the same point explicitly: information is information, not matter or energy [B12]. That clarity was lost in engineering practice. The bit became an object to which energy, entropy and location could be ascribed.

[S] The mechanism is not new in the history of science. Entropy, the electric field, the gene — all of these abstractions were at certain times treated as substantial things, with consequences for theory. In the case of information the hypostatisation leads to a false physics: if the bit itself had entropy, then erasing it would have to produce heat — independently of the carrier. That conclusion is compelling only for someone who has confused the abstraction with physical reality.

[B] This section is marked as a sketch because it offers an interpretation and provides no demonstration. The historical parallels are analogies; they support the thesis, they do not prove it.

.11 Conclusion

1. **[B]** Landauer's proof for the physical operation RESTORE TO ONE on a thermal equilibrium ensemble is correct and precise.
2. **[B]** It shows a lower bound for *carrier operations* — not a universal statement about abstract information.

3. **[B]** The transfer to the information-theoretic notion of the bit presupposes that every information-theoretic erasure is bound up with a state-space-shrinking carrier operation. Landauer does not prove this connection — and by Bennett it does not hold in general.
4. **[Q]** The double sense of "bit" is present in the original paper and encouraged the generalisation in the reception.
5. **[Q]** Landauer himself discusses the limits of his ensemble argument and concedes that non-random input data lower the dissipation below $kT \ln 2$.
6. **[B]** What is not shown here: that Landauer's principle is false. What is shown is that it has jurisdiction over a subject matter other than the one it is usually cited for.

.12 Outlook

[B] A formulation that reflects the actual proof reads:

> *The physical operation that puts a bistable system in thermal equilibrium into a definite state irrespective of its initial state requires at least $kT \ln 2$ of dissipation. If this system serves as an information carrier and the operation corresponds to erasing a bit, the bound applies to the associated carrier operation — not to information-theoretic erasure as such.*

[H] Open question. Under what conditions does a given computational process enforce a state-space-shrinking carrier operation? Bennett answers the limiting case (in principle never, if everything is carried along); the practically interesting question is quantitative: how much carrier state space must an architecture finally give up at a given memory depth? This is the same open question as the continuous entropy current in A271 — posed from the other side.

.13 Layer-status summary

Statement	Status
Definition of information (abstract, without energy)	**[SETZUNG]**
Definition of carrier (physical, with energy)	**[SETZUNG]**
Dictionary mapping A271 ↔ A272	**[SETZUNG]**
Proposition 1: energy attaches to the carrier, not to the bit	**[B]**
Bus example: bus width is hardware, not information	**[B]**
Landauer's double-well model and RESTORE TO ONE	**[Q]**
$\Delta S = k \ln 2$ presupposes $W{=}2 \to W{=}1$ at the carrier	**[Q]**
Proposition 2: interpretive erasure does not shrink W	**[B]**
The objection is a finding about jurisdiction, not a refutation	**[B]**
Bennett: logically irreversible functions executable reversibly	**[B]**
Landauer's ensemble restriction (original quotation)	**[Q]**
Landauer's own resolution for random data	**[Q]**
For correlated data the bound is kTH, not $kT \ln 2$	**[B]**
Landauer uses "bit" in a double sense	**[Q]**
Feedback bound $Q_{\mathrm{min}} = kT(\ln 2 - I)$, I in nats	**[Q]**
The knowledge I is itself carrier-bound (Bennett)	**[B]**
Reception findings: Lairez / Norton / Hemmo–Shenker	**[Q]**
Structural reasons for the persistence	**[S]**
Language table: carrier versus information	**[B]**

Statement	Status
Hypostatisation as an interpretive frame	**[S]**
Historical parallels (entropy, field, gene)	**[S]**
No demonstration that Landauer’s principle is false	**[B]**
Quantitative form of the unavoidable state-space relinquishment	**[H]**

Balance: 10× **[B]** · 8× **[Q]** · 3× **[SETZUNG]** · 3× **[S]** · 1× **[H]** = 25 entries.

.14 Use of AI tools

This document was produced with the aid of AI-assisted tools. The questions, the stipulations and all decisions of content are the author’s; the tools draft the prose, check the arithmetic and produce the verification scripts. Responsibility for content and for errors rests entirely with the author. The full wording is given in A010.

.15 Check script

This document carries no figures of its own; the numerical statements are held in A271 (`a271_landauer.py`) and A273 (`a273_rechenkugel.py`).

.16 What remains open

Under what conditions a given computational process enforces a state-space-shrinking carrier operation is quantitatively open; Bennett answers the limiting case, the practically interesting question remains.

.17 Supersedes

This document has no direct counterpart in the older corpus; it supplements A271 with the textual criticism and the reception history.

Sources

[B1] R. Landauer, *Irreversibility and Heat Generation in the Computing Process*, IBM J. Res. Dev. **5**, 183 (1961).

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[B3] J. Earman and J. D. Norton, *Exorcist XIV: The Wrath of Maxwell's Demon. Part II: From Szilard to Landauer and Beyond*, Stud. Hist. Phil. Mod. Phys. **30**, 1 (1999).

[B4] J. D. Norton, *All Shook Up: Fluctuations, Maxwell's Demon and the Thermodynamics of Computation*, Entropy **15**, 4432 (2013).

[B5] T. Sagawa and M. Ueda, *Nonequilibrium Thermodynamics of Feedback Control*, Phys. Rev. E **85**, 021104 (2012).

[B6] J. M. R. Parrondo, J. M. Horowitz and T. Sagawa, *Thermodynamics of Information*, Nature Physics **11**, 131 (2015).

[B7] J. Bub, *Maxwell's Demon and the Thermodynamics of Computation*, Stud. Hist. Phil. Mod. Phys. **32**, 569 (2001).

[B8] C. H. Bennett, *The Thermodynamics of Computation — a Review*, Int. J. Theor. Phys. **21**, 905 (1982).

[B9] *(incomplete — identifier to be supplied)* D. Lairez, *Landauer's Principle: A Critical Assessment*, arXiv preprint (2024).

[B10] M. Hemmo and O. Shenker, *The Multiple-Computations Theorem and the Physics of Singling Out a Computation*, The Monist **105**, 175 (2022).

[B11] C. E. Shannon, *A Mathematical Theory of Communication*, Bell System Technical Journal **27**, 379 (1948).

[B12] *(reconstructed — page number to be checked)* N. Wiener, *Cybernetics: or Control and Communication in the Animal and the Machine*, MIT Press, Cambridge, MA (1948).

Entries marked incomplete *or* reconstructed *were supplied from the citation context or carry an open placeholder; they are to be checked against the working files before publication.*